

Úloha 3.2

$$\langle f, g \rangle = \int_1^5 f(x)g(x)dx$$

a) Gram-Schmidtova ortogonalizace. Najdeme Ort. ~~span~~ ^{span} $\{1, x, x^2\}$.
Označme $v_1 = 1$ $v_2 = x$ $v_3 = x^2$

$$1) u_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{\int_1^5 1 \cdot 1 dx}} \cdot 1 = \frac{1}{\sqrt{[x]_1^5}} = \frac{1}{\sqrt{5-1}} = \frac{1}{2}$$

2) Najdeme Ob projekci v_2 na $\text{Span}\{\frac{1}{2}\}$.

$$w_2 = \langle u_1, v_2 \rangle \cdot u_1 = \langle \frac{1}{2}, x \rangle \cdot \frac{1}{2} = \left(\int_1^5 \frac{1}{2} x dx \right) \cdot \frac{1}{2} =$$

$$= 6 \cdot \frac{1}{2} = 3$$

$$u_2 = \frac{1}{\|v_2 - w_2\|} \cdot (v_2 - w_2) = \frac{1}{\sqrt{\int_1^5 (x-3)^2 dx}} \cdot (x-3) = \frac{1}{\frac{4}{\sqrt{5}}} (x-3) = \frac{\sqrt{5}}{4} (x-3)$$

3) Najdeme Ob projekci v_3 na $\text{Span}\{\frac{1}{2}, \frac{\sqrt{3}}{4}(x-3)\}$

$$w_3 = \langle \frac{1}{2}, x^2 \rangle \frac{1}{2} + \langle \frac{\sqrt{3}}{4}(x-3), x^2 \rangle \cdot \frac{\sqrt{3}}{4}(x-3) =$$

$$= \left(\int_1^5 \frac{1}{2} x^2 dx \right) \cdot \frac{1}{2} + \left(\int_1^5 \frac{\sqrt{3}}{4} (x-3) \cdot x^2 dx \right) \cdot \frac{\sqrt{3}}{4} (x-3) =$$

$$= \frac{62}{3} \cdot \frac{1}{2} + 8\sqrt{3} \cdot \frac{\sqrt{3}}{4} (x-3) = \frac{31}{3} + \frac{8 \cdot 3}{4} (x-3) =$$

$$= \frac{31}{3} + 6x - 18 = -\frac{23}{3} + 6x$$

$$u_3 = \frac{1}{\|v_3 - w_3\|} \cdot (v_3 - w_3) = \frac{1}{\sqrt{\int_1^5 (x^2 - 6x + \frac{23}{3})^2 dx}} \left(x^2 - 6x + \frac{23}{3} \right) = \frac{3\sqrt{5}}{16} \left(x^2 - 6x + \frac{23}{3} \right)$$

Tím pádem OG bŕze prostoru $\text{span}\{1, x, x^2\}$ je

$$B = \left(\frac{1}{2}, \frac{\sqrt{3}}{4}(x-3), \frac{3\sqrt{5}}{16}(x^2-6x+\frac{23}{3}) \right).$$

⑥ ~~Viz~~ Ortogonální projekce funkce $\sin x$ na tento podprostor má tvar.

$$\begin{aligned} W_{\sin x} &= \langle u_1, \sin x \rangle u_1 + \langle u_2, \sin x \rangle u_2 + \langle u_3, \sin x \rangle u_3 = \\ &= \left(\int_1^5 \frac{1}{2} \sin x \, dx \right) \cdot \frac{1}{2} + \left(\int_1^5 \frac{\sqrt{3}}{4} (x-3) \sin x \, dx \right) \cdot \frac{\sqrt{3}}{4} (x-3) + \\ &+ \left(\int_1^5 \frac{3\sqrt{5}}{16} (x^2-6x+\frac{23}{3}) \sin x \, dx \right) \cdot \frac{3\sqrt{5}}{16} (x^2-6x+\frac{23}{3}) \approx \\ &\approx 0,12 \cdot \frac{1}{2} + (-1,5) \cdot \frac{\sqrt{3}}{4} (x-3) + (-0,14) \cdot \frac{3\sqrt{5}}{16} (x^2-6x+\frac{23}{3}) = \\ &\approx -0,05x^2 - 0,348x + 1,623 \end{aligned}$$

$$\begin{aligned} \textcircled{c} \text{ Norma rozdílu } \|\sin(x) - (ax^2+bx+c)\| &= \\ &= \sqrt{\langle (\sin x) + 0,05x^2 + 0,348x - 1,623, \sin x + 0,05x^2 + 0,348x - 1,623 \rangle} = \\ &= \sqrt{\int_1^5 (\sin x + 0,05x^2 + 0,348x - 1,623)^2 \, dx} \approx 0,32 \end{aligned}$$